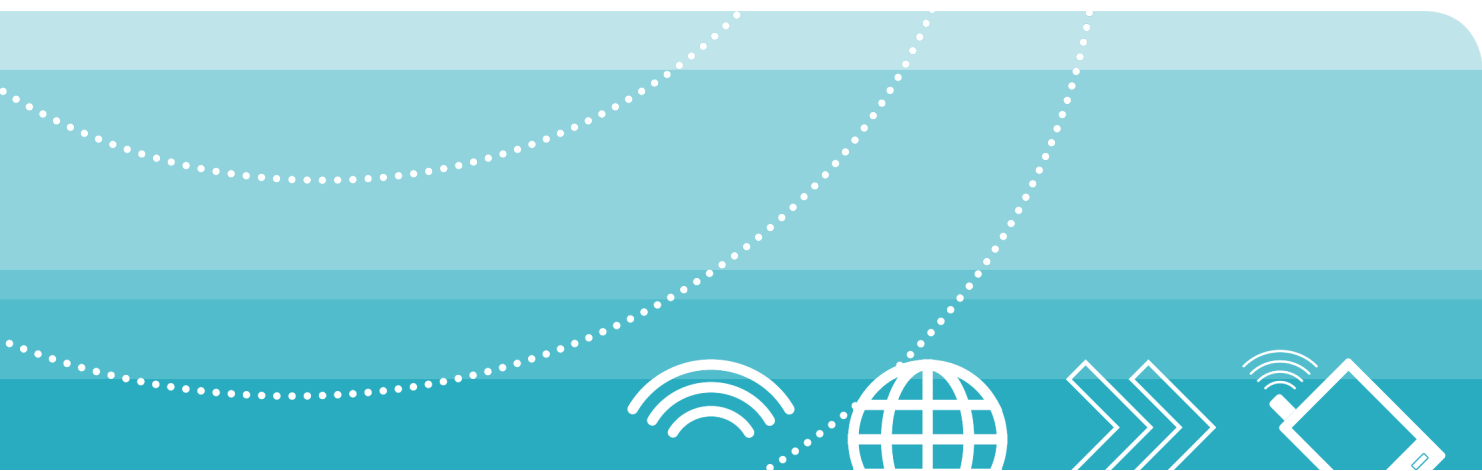




MT5710-CN 5G RedCap 模组 Linux 内核驱动集成指导

V100R001C00

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1 关于本文档

修订记录

文档版本	发布日期	说明
01	2023-06-09	第一次发布

2 目的

本文档主要是针对MT571X系列模组，基于Linux内核系统（如Android、Ubuntu和Chrome OS等）驱动集成开发活动进行相关的指导说明，主要面向基于Linux内核产品开发商的驱动开发人员。

3 范围

功能	描述	
支持的系统	Linux、Android和Chrome OS	
支持的模块制式	LTE/NR	
支持的内核版本	2.6.12以前的版本	本文档不支持
	2.6.12 ~ 2.6.21	支持USB串口驱动集成（ 章节5 USB串口驱动集成 ）
	2.6.22 ~ 2.6.31	- 支持USB串口驱动集成（章节5 USB串口驱动集成） - 支持CDC ECM驱动集成（章节6 CDC ECM驱动集成）
	2.6.32及以后的版本	- 支持USB串口驱动集成（章节5 USB串口驱动集成） - 支持CDC ECM驱动集成（章节6 CDC ECM驱动集成）

4 概要

本文档主要介绍如何修改Linux内核代码，使客户系统（如Android等）能够正常加载模块。因本文档支持的系统都是基于Linux内核的系统，下文统一称Linux侧，基于Linux内核版本5.15.1进行修改并以此版本作为内核进行运行。

MT571X在Linux侧使用的驱动需要进行内核编译配置项设置，确保驱动编译进内核。

本文中linux_src代表所下载的内核源代码根目录。

5 USB 串口驱动集成

5.1 简要说明

5.2 代码修改

5.1 简要说明

MT5710商用版本默认接口，包括：PCUI、Diag、ECM、NCM等。

- 各端口作用详见：MT5710-CN 5G RedCap模组USB接口应用指南

5.2 代码修改

这部分集成涉及修改的Linux内核源码文件为：

Linux_src/drivers/usb/serial/option.c

5.2.1 增加宏定义

步骤1：在linux_src/drivers/usb/serial/option.c文件中添加如下宏

```
#define TDTECH_VENDOR_ID 0x3466
```

步骤2：在linux_src/drivers/usb/serial/option.c文件中的static const struct usb_device_id option_ids[]的id列表中增加如下语句,用于匹配设备数据。

```
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x03) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x06) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x0A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x0B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x0E) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x10) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x12) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x13) },
```

```
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x14) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x15) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x18) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x19) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x1A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x1B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x1C) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x01, 0x1D) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x03) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x06) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x0A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x0B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x0E) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x10) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x12) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x13) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x14) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x15) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x18) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x19) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x1A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x1B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x1C) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x02, 0x1D) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x03) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x06) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x0A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x0B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x0E) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x10) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x12) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x13) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x14) },
```

```
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x15) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x18) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x19) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x1A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x1B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x1C) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x03, 0x1D) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x03) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x06) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x0A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x0B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x0E) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x10) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x12) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x13) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x14) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x15) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x18) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x19) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x1A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x1B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x1C) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x04, 0x1D) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x03) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x06) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x0A) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x0B) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x0E) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x10) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x12) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x13) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x14) },  
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x15) },
```

```
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x18) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x19) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x1A) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x1B) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x1C) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x05, 0x1D) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x03) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x06) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x0A) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x0B) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x0E) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x10) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x12) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x13) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x14) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x15) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x18) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x19) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x1A) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x1B) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x1C) },
{ USB_VENDOR_AND_INTERFACE_INFO(TDTECH_VENDOR_ID, 0xff, 0x06, 0x1D) },
```

5.2.2 内核的编译配置

文本配置方式，修改linux_src/目录下的.config文件。

- USB串口驱动相关的配置项：
CONFIG_USB_SERIAL=y
CONFIG_USB_SERIAL_OPTION=y
CONFIG_USB_SERIAL_WWAN=y

说明

USB串口驱动相关配置项在make menuconfig的图像界面配置方法，请参考[8.1](#)。

6 CDC ECM 驱动集成

6.1 简要说明

6.2 内核编译配置

6.1 简要说明

CDC ECM驱动是MT571X模块适配标准的ECM通用驱动，无需额外的代码修改，直接配置编译项即可。该功能支持内核2.6.22及以后的版本。

6.2 内核编译配置

文本配置方式，修改Linux内核的编译配置（在`linux_src/`目录下的`.config`文件中）。

CDC ECM驱动的相关配置项：

`CONFIG_USB_USBNET=y`

`CONFIG_USB_NET_CDCETHER=y`

`CONFIG_NETDEVICES=y`

下面这个参数适用于arm架构的PC，编译arm架构的时候请加上这个参数（较少场景使用）：

`CONFIG_USB_CONFIGFS_ECM=y`

说明

make menuconfig的图像界面配置，请参考[8.3 CDC NCM驱动编译配置项](#)。

7 NCM 驱动集成

7.1 简要说明

7.2 内核编译配置

7.1 简要说明

NCM驱动是MT571X模块适配标准的NCM通用驱动，无需额外的代码修改，直接配置编译项即可。该功能支持内核2.6.22及以后的版本。

7.2 内核编译配置

文本配置方式，修改Linux内核的编译配置（在`linux_src/`目录下的`.config`文件中）。

NCM驱动的相关配置项：

`CONFIG_USB_USBNET=y`

`CONFIG_NETDEVICES=y`

下面这个参数适用于arm架构的PC，编译arm架构的时候请加上这个参数（较少场景使用）：

`CONFIG_USB_CONFIGFS_NCM=y`

说明

make menuconfig的图像界面配置，请参考[8.3 CDC NCM驱动编译配置项](#)。

8 make menuconfig 配置

界面编译方式适用于不在 [.config] 文件中加入配置的方式，与章节5.2.2、章节6.2、章节7.2是重复的操作。

8.1 USB串口驱动编译配置项

8.2 CDC ECM 驱动编译配置项

8.3 CDC NCM驱动编译配置项

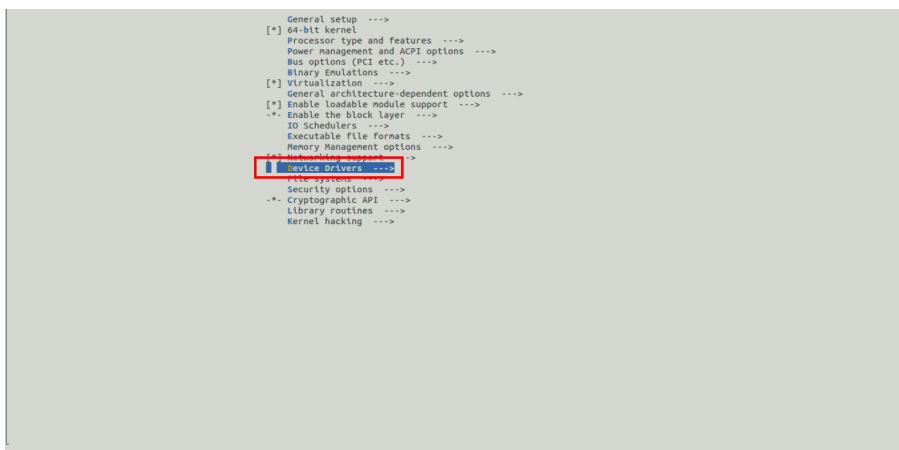
8.1 USB 串口驱动编译配置项

在使用界面配置之前，要保证鼎桥的信息写入相应的代码中，具体操作请阅读[5 USB 串口驱动集成](#)。

/usr/src/linux_src/ 目录下执行命令 “make menuconfig”

```
root@tdtech-Produ:/usr/src/linux-5.15.1# make menuconfig
```

出现界面中进行如下配置：



```

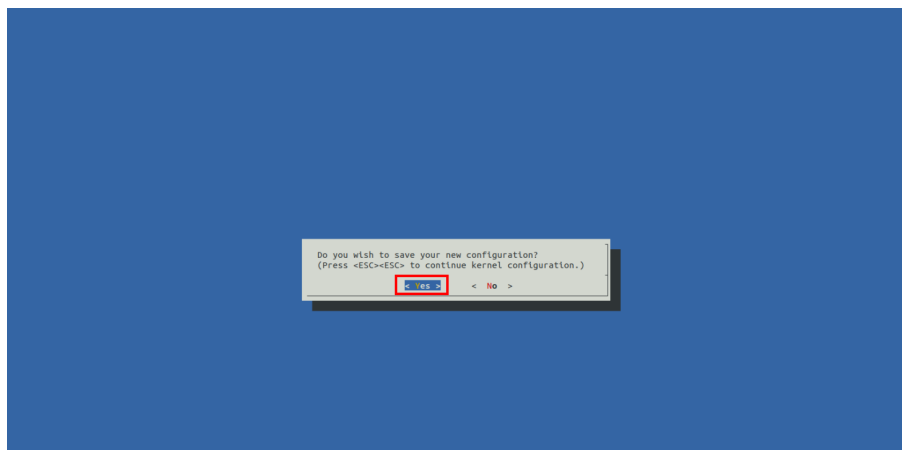
<+> Hardware Monitoring support --->
<+> Thermal drivers --->
[*] Watchdog Timer Support --->
<+> Sonics Silicon Backplane support --->
<+> Broadcom specific AMBA --->
<+> Multifunction device drivers --->
<+> Voltage and Current Regulator Support --->
<+> Remote Controller support --->
<+> HDMI CEC drivers --->
<+> Multimedia support --->
<+> Graphics support --->
<+> Sound card support --->
[*] USB support --->
<+> USB HID support --->
<+> Sony MemoryStick card support --->
<+> LED Support --->
<+> Accessibility support --->
<+> Infrared support --->
[*] Real Time Clock --->
[*] DMA Engine support --->
<+> DMABUF options --->
<+> Auxiliary display support --->
<+> Userspace I/O drivers --->
<+> VFIO Non-Privileged userspace driver framework --->
<+> Virtualization drivers --->
[*] Virtio drivers --->
<+> vDPA drivers --->
[*] VHOST drivers --->
<+> Microsoft Hyper-V guest support --->
<+> Greybus support --->
<+> Data acquisition support (coned1) --->
<+> Staging drivers --->
<+> X86 Platform Specific Device Drivers --->
<+> Platform support for Goldfish virtual devices --->
<+> Platform support for Chrome hardware --->
<+> Platform support for Mellanox hardware --->
<+> Microsoft Surface Platform-Specific Device Drivers --->
<+> Common Clock Framework --->
<+> Hardware Spinlock drivers --->
v(+)

<+> UHCI HCD (most Intel and VIA) support
<+> SLB11HS HCD support
<+> R8A66597 HCD support
[ ] HCD test mode support
*** USB Device Class drivers ***
<+> USB Modem (CDC ACM) support
<+> USB Printer support
<+> USB Wireless Device Management support
<+> USB Test and Measurement Class support
*** NOTE: USB_STORAGE depends on SCSI but BLK_DEV_SD may ***
*** also be needed; see USB_STORAGE help for more info ***
<+> USB Mass Storage support
[ ] USB Mass Storage Verbose debug
<+> Realtek Card Reader support
<+> Datafab Compact Flash Reader support
<+> Freecom USB/ATAPI Bridge support
<+> ISO-200 USB/ATA Bridge support
<+> USBAT/USBAT02-based storage support
<+> Sandisk SDDR-09 (and other SmartMedia, including DPCH) support
<+> Sandisk SDDR-55 SmartMedia support
<+> Lexar Jumpshot Compact Flash Reader
<+> Olympus MAUSB-10/Fuji DPC-R1 support
<+> Support OneTouch button on Raptor Hard Drives
<+> Support for Rio Karma music player
<+> SAT emulation on Cypress USB/ATA Bridge with ATACB
<+> USB ENE card reader support
<+> USB Attached SCSI
*** USB Imaging devices ***
<+> USB Muxtek HOCBB Digital Camera support
<+> Microtek X60USB scanner support
<+> USB/IIP support
<+> Cadence USB Support
<+> Inventra Highspeed Dual Role Controller
<+> DesignWare USB2 DRD Core Support
<+> DesignWare USB2 DRD Core Support
<+> Chipidea Highspeed Dual Role Controller
<+> NXP ISP 1760/1761/1763 support
*** USB port drivers ***
[*] USB Serial Converter support --->
*** USB Miscellaneous drivers ***
v(+)

--- USB Serial Converter support
[ ] USB Serial Console device support (NEW)
[*] USB Generic Serial Driver
<+> USB Serial Converter Driver (NEW)
<+> USB Aircable Bluetooth Dongle Driver (NEW)
<+> USB ARK Micro 3116 USB Serial Driver (NEW)
<+> USB Belkin and Peracom Single Port Serial Driver (NEW)
<+> USB Winchiphead CH341 Single Port Serial Driver (NEW)
<+> USB ConnectTech WhiteHEAT Serial Driver (NEW)
<+> USB Digi International Accleport USB Serial Driver (NEW)
<+> USB CP210x Family of UART Bridge Controllers (NEW)
<+> USB Cypress MB USB Serial Driver (NEW)
<+> USB Enpeg enpeg-car Mark I/II Driver (NEW)
<+> USB FTDI Single Port Serial Driver (NEW)
<+> USB Handspring Visor / Palm m50x / Sony Clie Driver (NEW)
<+> USB PocketPC PDA Driver (NEW)
<+> USB IP Dongle Serial Driver (NEW)
<+> USB Inside Out Edgeport Serial Driver (NEW)
<+> USB Fintek FB1232 Single Port Serial Driver (NEW)
<+> USB Fintek FB1532/534 Multi-Ports Serial Driver (NEW)
<+> USB Garmin GPS driver (NEW)
<+> USB IPWireless (3C UMTS TDD) Driver (NEW)
<+> USB Infinity USB Unlimited Phoenix Driver (NEW)
<+> USB Keyspan PDA / Xircom Single Port Serial Driver (NEW)
<+> USB Keyspan USA-xxx Serial Driver (NEW)
<+> USB KLSKUSB105 (Palmconnect) Driver (NEW)
<+> USB KOBIL chipcard reader (NEW)
<+> USB MCT Single Port Serial Driver (NEW)
<+> USB Metrologic Instruments USB-POS Barcode Scanner Driver (NEW)
<+> USB Moschip 7720 Serial Driver (NEW)
<+> USB Moschip 7840/7820 USB Serial Driver (NEW)
<+> USB Moxa UPORT Serial Driver (NEW)
<+> USB Navman GPS device (NEW)
<+> USB Prolific 2303 Single Port Serial Driver (NEW)
<+> USB Ours Technology Inc. OTI-6858 USB To RS232 Bridge Controller (NEW)
<+> USB Qualcomm Auxiliary Serial Driver (NEW)
<+> USB Qualcomm modem (NEW)
<+> USB SPCPBx5 USB To Serial Driver (NEW)
<+> USB Safe Serial (Encapsulated) Driver (NEW)
v(+)

```

保存退出，自动更新 .config 文件



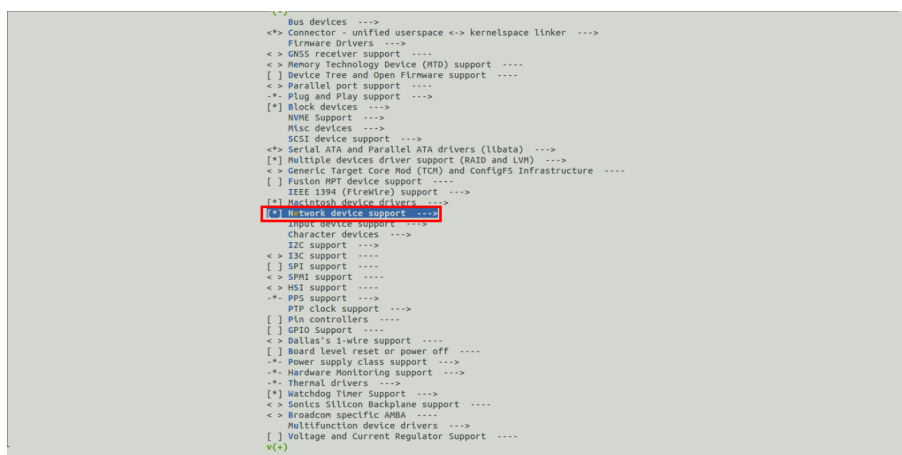
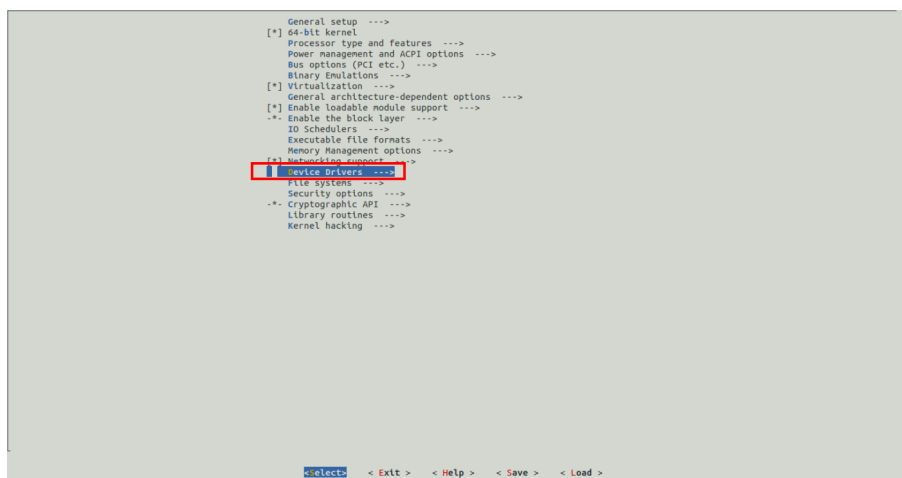
/usr/src/linux_src/ 目录下执行 make 进行编译

8.2 CDC ECM 驱动编译配置项

/usr/src/linux_src/ 目录下执行命令 “make menuconfig”

```
root@tdtech-Produ:/usr/src/linux-5.15.1# make menuconfig
```

出现界面中进行如下配置：



```

--- Network device support
[*] Network core driver support
< > Bonding driver support
< > Dummy net driver support
< > WireGuard secure network tunnel
< > EQL (serial line load balancing) support
[ ] Fibre Channel driver support
< > Intermediate Functional Block support
< > Ethernet team driver support ----
< > MAC-VLAN support
< > IP-VLAN support
< > Virtual extensible Local Area Network (VXLAN)
< > Generic Network Virtualization Encapsulation
< > Bare UDP Encapsulation
< > GPRS Tunneling Protocol datapath (GTP-U)
< > IEEE 802.1AE MAC-level encryption (MACsec)
<+> Network console logging support
< > Universal TUN/TAP device driver support
[ ] Support for cross-endian vnet headers on little-endian kernels
< > Virtual ethernet pair device
< > Virtual netlink monitoring device
< > ARCNet support ----
[*] Ethernet driver support ---
< > FDDI driver support
[ ] HIPPI driver support
< > General Instruments Surfbord 1000
<+> PHY Device support and Infrastructure ---
<+> RDIO bus device drivers ---
< > PCS device drivers ----
< > PPP (point-to-point protocol) support
< > SLIP (serial line) support
<+> USB Network Adapters ---
[*] Wireless LAN ----
[ ] Man Interfaces support ----
Wireless LAN ----
< > VMware VMXNET3 ethernet driver
< > FUJITSU Extended Socket Network Device driver
< > Simulated networking device
< > Fallover driver
[ ] ISDN support ----

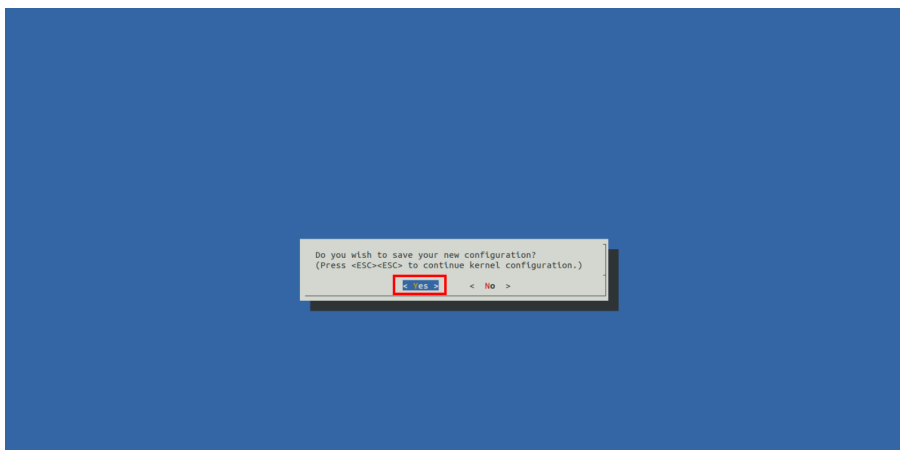
```

```

--- USB Network Adapters
< > USB CATC NetHate-based Ethernet device support (NEW)
< > USB KLSI KLSUS001-based ethernet device support (NEW)
< > USB Pegasus/Pegasus-II based ethernet device support (NEW)
< > USB RTL8150 based ethernet device support (NEW)
< > Realtek RTL8152/RTL8153 Based USB Ethernet Adapters (NEW)
< > Microchip LAN78XX Based USB Ethernet Adapters (NEW)
<+> Multi-purpose USB Networking Framework
<+> ASIX AX88179/178A USB 3.0/2.0 to Gigabit Ethernet (NEW)
<+> ASIX AX88179/178A USB 3.0/2.0 to Gigabit Ethernet (NEW)
<+> CDC Ethernet support (smart devices such as cable modems)
<+> CDC Ethenet support (NEW)
<+> CDC NCM support (NEW)
< > Huawei NCM embedded AT channel support (NEW)
< > CDC NCM support (NEW)
< > Davicom DM96xx based USB 10/100 ethernet devices (NEW)
< > Corechip-sz SR9700 based USB 1.1 10/100 ethernet devices (NEW)
< > Corechip-sz SR9800 based USB 2.0 10/100 ethernet devices (NEW)
< > SMSC LAN75XX based USB 2.0 gigabit ethernet devices (NEW)
< > SMSC LAN95XX based USB 2.0 10/100 ethernet devices (NEW)
< > Genesys GL620USB-A based cables (NEW)
< > Netchip 1080 based cables (LapLink, ...) (NEW)
< > Prolific PL-2301/2302/25A1/27A1 based cables (NEW)
< > Hoshich HCS7830 based Ethernet adapters (NEW)
< > Host for RNDIS and ActiveSync devices (NEW)
<+> Simple USB Network Links (CDC Ethernet subset) (NEW)
[ ] All RS485 based "USB 2.0 Data Link" cables (NEW)
[ ] AnchorChips 2720 based cables (Xlrcon PGUNET, ...) (NEW)
[*] eTEK based host-to-host cables (Advance, Belkin, ...) (NEW)
[*] Embedded ADN Linux Links (Vpaq, ...) (NEW)
[ ] Epson 2888 based firmware (DEVELOPMENT) (NEW)
[ ] KT Technology KC2190 based cables (Instalet) (NEW)
<+> Sharp Zaurus (stock ROMs) and compatible (NEW)
< > Conexant CX82310 USB ethernet port (NEW)
< > Samsung Kalnia based LTE USB modem (NEW)
< > QMI WMAN driver for Qualcomm MSM based 3G and LTE modems (NEW)
< > Option USB High Speed Mobile Devices (NEW)
< > Intellon PLC based usb adapter (NEW)
< > Apple iPhone USB Ethernet driver (NEW)
< > USB-to-WMAN Driver for Sierra Wireless modems (NEW)
v(+)

```

保存退出，自动更新 .config 文件



/usr/src/linux_src/ 目录下执行 make 进行编译

8.3 CDC NCM 驱动编译配置项

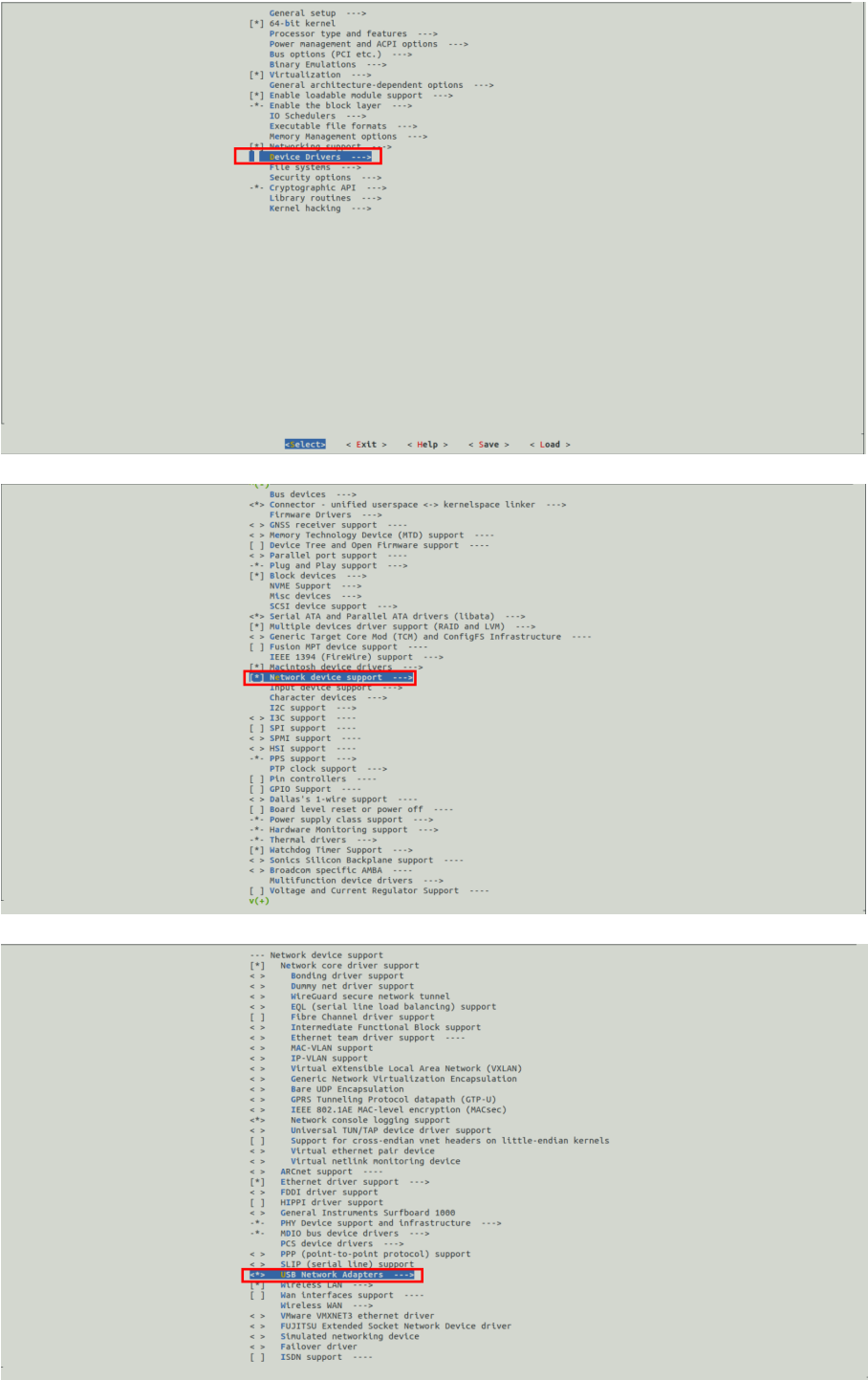
/usr/src/linux_src/ 目录下执行命令 “make menuconfig”

```

root@tdtech-Produ:/usr/src/linux-5.15.1# make menuconfig

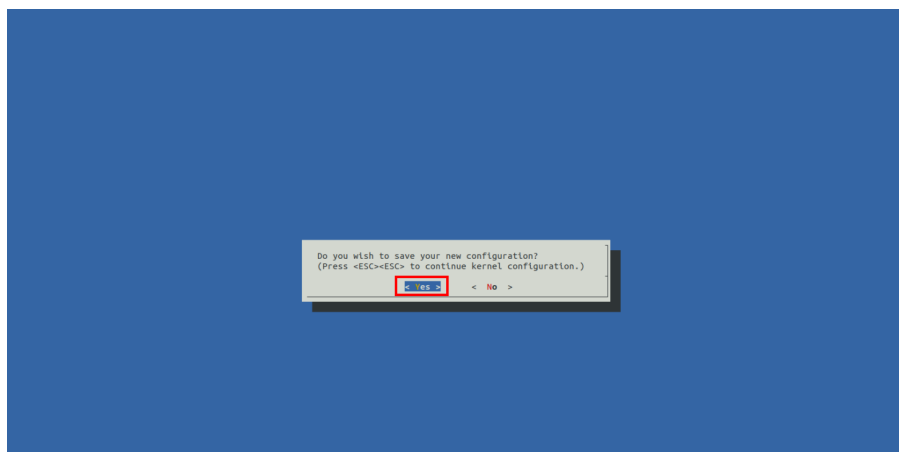
```

出现界面中进行如下配置：





保存退出，自动更新 .config 文件



/usr/src/linux_src/ 目录下执行 make 进行编译

9 拨号分配 IP

9.1 以NCM举例说明

9.1 以 NCM 举例说明

查看启动信息：dmesg 可以看到加载的VID/PID；枚举出来的端口ttyUSB0，ttyUSB1，ttyUSB2，ttyUSB3；加载的网卡usb0(或者其他网卡名字)信息。

```
root@lot2-Lenovo-Product:/dev# dmesg
[102843.703681] usb 3-10: new high-speed USB device number 20 using xhci_hcd
[102843.911935] usb 3-10: config 2 interface 1 altsetting 1 has 2 endpoint descriptors, different from the interface descriptor's value: 3
[102843.912345] usb 3-10: new USB device found, idVendor=3466, idProduct=3381, bcdDevice= 1.02
[102843.912353] usb 3-10: New USB device strings: Mfr=13, Product=14, SerialNumber=15
[102843.912358] usb 3-10: Product: TDTECH MT571X
[102843.912363] usb 3-10: Manufacturer: TD Tech Ltd.
[102843.912367] usb 3-10: SerialNumber: 0123456789ABCDEF
[102843.991620] cdc ncm 3-10:2.2: MAC-Address: 0c:5b:8f:27:9a:64
[102844.000196] cdc ncm 3-10:2.0 usb0: register 'cdc ncm' at usb-0000:00:14.0-10, CDC NCM, 0c:5b:8f:27:9a:64
[102844.007921] option 3-10:2.2: GSM modem (1-port) converter detected
[102844.008008] usb 3-10: GSM modem (1-port) converter now attached to ttyUSB0
[102844.008172] option 3-10:2.3: GSM modem (1-port) converter detected
[102844.008233] usb 3-10: GSM modem (1-port) converter now attached to ttyUSB1
[102844.008406] option 3-10:2.4: GSM modem (1-port) converter detected
[102844.008461] usb 3-10: GSM modem (1-port) converter now attached to ttyUSB2
[102844.008614] option 3-10:2.5: GSM modem (1-port) converter detected
[102844.008679] usb 3-10: GSM modem (1-port) converter now attached to ttyUSB3
root@lot2-Lenovo-Product:/dev#
```

检查端口: `ls -all /dev/ttyUSB*`

```
root@iot2-Lenovo-Product:/dev# ls -all /dev/ttyUSB*
crw-rw---- 1 root dialout 188, 0 6月 9 16:15 /dev/ttyUSB0
crw-rw---- 1 root dialout 188, 1 6月 9 16:15 /dev/ttyUSB1
crw-rw---- 1 root dialout 188, 2 6月 9 16:15 /dev/ttyUSB2
crw-rw---- 1 root dialout 188, 3 6月 9 16:15 /dev/ttyUSB3
root@iot2-Lenovo-Product:/dev#
```

赋予端口权限: `chmod 777 /dev/ttyUSB*`

```
root@iot2-Lenovo-Product:/dev# chmod 777 /dev/ttyUSB*
```

验证PCUI口正常:

开启双**Terminal**，一个输出打印结果：cat /dev/ttyUSB1，一个发送AT指令：echo -e "ATE1\r\n" > /dev/ttyUSB1

[illegible]

查看网卡: ifconfig

```
root@iot2-Lenovo-Product:/home/iot2# ifconfig
eth0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 00:e0:4c:97:02:a2 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 17

eth1: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether fc:4d:d4:f5:00:d6 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 20 memory 0xf7d00000-f7d20000

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 842 bytes 61136 (61.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 842 bytes 61136 (61.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

usb0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 0c:5b:8f:27:9a:64 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

如果查不到网口usb0请执行命令:

```
ifconfig -a
```

```
ifconfig usb0 up
```

拨号: `echo -e "AT^NDISDUP=1,1\r\n" > /dev/ttyUSB1`

```
root@iot2-Lenovo-Product:/home/iot2# echo -e "at+ndisdup=1,1\n\r" > /dev/ttyUSB1
^DSAMBR: 1,1000000,100000,"CMNET.MNC000.MCC460.GPRS",
OK
^NDISSTAT: 1,,,"IPV4"
^NDISSTAT: 1,,,"IPV6"
```

拨号之后请查看网卡状态(usb0已RUNNING): ifconfig 【注: 如果没有获取到IP, 输入命令: dhclient -d usb0】

```
root@iot2-Lenovo-Product:/home/iot2# ifconfig
eth0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether fc:4d:d4:f5:00:d6 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 17

eth1: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether fd:d4:d4:f5:00:d6 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 20 memory 0xf7d00000-f7d20000

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 2207 bytes 165007 (165.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2207 bytes 165007 (165.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

usb0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.33.91.82 netmask 255.0.0.0 broadcast 10.255.255.255
    ether fe80::3b49:e7cc:8055:193c prefixlen 64 scopeid 0x20<link>
    inet6 2409:8900:2769:29bb:31f2:b2d4:aee7:88f0 prefixlen 64 scopeid 0x0<global>
    inet6 2409:8900:2769:29bb:e994:ad3e:6536:63af prefixlen 64 scopeid 0x0<global>
    ether 0c:5b:8f:27:9a:64 txqueuelen 1000 (Ethernet)
    RX packets 184 bytes 77669 (77.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 281 bytes 30907 (30.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

测试ping 百度: ping www.baidu.com

```
root@iot2-Lenovo-Product:/home/iot2# ping www.baidu.com
PING www.a.shifen.com (39.156.66.18) 56(84) bytes of data.
64 bytes from 39.156.66.18: icmp_seq=1 ttl=51 time=113 ms
64 bytes from 39.156.66.18: icmp_seq=3 ttl=51 time=46.8 ms
64 bytes from 39.156.66.18: icmp_seq=4 ttl=51 time=349 ms
64 bytes from 39.156.66.18: icmp_seq=5 ttl=51 time=516 ms
64 bytes from 39.156.66.18: icmp_seq=6 ttl=51 time=95.5 ms
64 bytes from 39.156.66.18: icmp_seq=7 ttl=51 time=337 ms
64 bytes from 39.156.66.18: icmp_seq=8 ttl=51 time=36.7 ms
```

10 FAQ

- 10.1 如何确认系统中是否已经存在正确的USB串口驱动
- 10.2 如何确认系统中是否已经存在正确的CDC ECM驱动
- 10.3 如何获取单板当前的端口映射情况信息
- 10.4 出现端口类似不是从ttyUSB0开始的情况如何处理
- 10.5 如何手动加载串口驱动
- 10.6 若无法映射端口或无法查找对应端口形态需要提供哪些log

10.1 如何确认系统中是否已经存在正确的 USB 串口驱动

打开Terminal，执行命令**dmesg**，查看内核的log信息。如查看到类似如下图内容，即说明设备正常加载。

```
root@iot2-Lenovo-Product:/dev# dmesg
[102843.763010] usb 3-10: new high-speed USB device number 20 using xhci_hcd
[102843.911935] usb 3-10: config 2 interface 1 altsetting 1 has 2 endpoint descriptors, different from the interface descriptor's value: 3
[102843.912345] usb 3-10: New USB device found, idVendor=3466, idProduct=3301, bcdDevice= 1.02
[102843.912353] usb 3-10: New USB device strings: Mfr=13, Product=14, SerialNumber=15
[102843.912358] usb 3-10: Product: TDTECH MT571X
[102843.912363] usb 3-10: Manufacturer: TD Tech Ltd.
[102843.912367] usb 3-10: SerialNumber: 0123456789ABCDEF
[102843.991020] cdc_ncm 3-10:2.0: MAC Address: 0c:5b:8f:27:9a:64
[102844.000100] cdc_ncm 3-10:2.0 usb0: register 'cdc_ncm' at usb-0000:00:14.0-10, CDC NCM, 0c:5b:8f:27:9a:64
[102844.007921] option 3-10:2.2: GSM modem (1-port) converter detected
[102844.008000] usb 3-10: GSM modem (1-port) converter now attached to ttyUSB0
[102844.008172] option 3-10:2.3: GSM modem (1-port) converter detected
[102844.008233] usb 3-10: GSM modem (1-port) converter now attached to ttyUSB1
[102844.008400] option 3-10:2.4: GSM modem (1-port) converter detected
[102844.008461] usb 3-10: GSM modem (1-port) converter now attached to ttyUSB2
[102844.008614] option 3-10:2.5: GSM modem (1-port) converter detected
[102844.008679] usb 3-10: GSM modem (1-port) converter now attached to ttyUSB3
root@iot2-Lenovo-Product:/dev#
```

10.2 如何确认系统中是否已经存在正确的 CDC ECM 驱动

打开Terminal，执行命令**dmesg**。如出现下图红框部分信息，则说明存在正确的CDC ECM驱动。

```
root@iot2-Lenovo-Product:/home/iot2# dmesg
[ 2477.016153] usb 3-4: new high-speed USB device number 8 using xhci_hcd
[ 2477.165852] usb 3-4: New USB device found, idVendor=3466, idProduct=3301, bcdDevice= 1.02
[ 2477.165867] usb 3-4: New USB device strings: Mfr=13, Product=14, SerialNumber=15
[ 2477.165873] usb 3-4: Product: TDTECH MT571X
[ 2477.165878] usb 3-4: Manufacturer: TD Tech Ltd.
[ 2477.165882] usb 3-4: SerialNumber: 0123456789ABCDEF
[ 2477.229800] cdc_ether 3-4:2.0 eth2: register 'cdc_ether' at usb-0000:00:14.0-4, CDC Ethernet Device, 0c:5b:8f:27:9a:64
[ 2477.243076] option 3-4:2.2: GSM modem (1-port) converter detected
[ 2477.243145] usb 3-4: GSM modem (1-port) converter now attached to ttyUSB0
[ 2477.251212] option 3-4:2.3: GSM modem (1-port) converter detected
[ 2477.251283] usb 3-4: GSM modem (1-port) converter now attached to ttyUSB1
[ 2477.258809] option 3-4:2.4: GSM modem (1-port) converter detected
[ 2477.258939] usb 3-4: GSM modem (1-port) converter now attached to ttyUSB2
[ 2477.259107] option 3-4:2.5: GSM modem (1-port) converter detected
[ 2477.259201] usb 3-4: GSM modem (1-port) converter now attached to ttyUSB3
```


10.3 如何获取单板当前的端口映射情况信息

(1) 执行命令dmesg，查看驱动是否加载成功。假如log信息中存在下图中红框部分信息，则说明驱动已经被成功加载（idProduct因产品的不同而不同）。

```
usb 3-10: new high-speed USB device number 20 using xhci_hcd
usb 3-10: config 2 interface 1 altsetting 1 has 2 endpoint descriptors, different from the int
usb 3-10: New USB device found, idVendor=3466, idProduct=3301, bcdDevice= 1.02
usb 3-10: New USB device strings: Mfr=13, Product=14, SerialNumber=15
usb 3-10: Product: TDTECH MT571X
usb 3-10: Manufacturer: TD Tech Ltd.
usb 3-10: SerialNumber: 0123456789ABCDEF
cdc_ncm 3-10:2.0: MAC-Address: 0c:5b:8f:27:9a:64
cdc_ncm 3-10:2.0 usb0: register 'cdc_ncm' at usb-0000:00:14.0-10, CDC NCM, 0c:5b:8f:27:9a:64
option 3-10:2.2: GSM modem (1-port) converter detected
usb 3-10: GSM modem (1-port) converter now attached to ttyUSB0
option 3-10:2.3: GSM modem (1-port) converter detected
usb 3-10: GSM modem (1-port) converter now attached to ttyUSB1
option 3-10:2.4: GSM modem (1-port) converter detected
usb 3-10: GSM modem (1-port) converter now attached to ttyUSB2
option 3-10:2.5: GSM modem (1-port) converter detected
usb 3-10: GSM modem (1-port) converter now attached to ttyUSB3
Product: /dev#
```

(2) 当前模块设备的Modem和PCUI等端口的设备文件名称查询命令：ls -all /dev/ttyUSB*

```
root@iot2-Lenovo-Product:/dev# ls -all /dev/ttyUSB*
crw-rw---- 1 root dialout 188, 0 6月 9 16:15 /dev/ttyUSB0
crw-rw---- 1 root dialout 188, 1 6月 9 16:15 /dev/ttyUSB1
crw-rw---- 1 root dialout 188, 2 6月 9 16:15 /dev/ttyUSB2
crw-rw---- 1 root dialout 188, 3 6月 9 16:15 /dev/ttyUSB3
root@iot2-Lenovo-Product:/dev#
```

(3) 对MT571X-CN模块，默认端口的对应关系举例如下：

Application Interface	/dev/ttyUSB0
PCUI	/dev/ttyUSB1
SerialC	/dev/ttyUSB2
GPS	/dev/ttyUSB3

具体实际端口与ttyUSBx对应关系，建议至/sys/bus/usb/drivers/option/XXXXX/interface具体确认。

```
root@iot2-Lenovo-Product:/sys/bus/usb/drivers/option#
root@iot2-Lenovo-Product:/sys/bus/usb/drivers/option# cat 3-10:2.2/interface
TDTECH Connect - Application Interface
root@iot2-Lenovo-Product:/sys/bus/usb/drivers/option# cat 3-10:2.3/interface
TDTECH Connect - PC UI Interface
root@iot2-Lenovo-Product:/sys/bus/usb/drivers/option# cat 3-10:2.4/interface
TDTECH Connect - SerialC
root@iot2-Lenovo-Product:/sys/bus/usb/drivers/option# cat 3-10:2.5/interface
TDTECH Connect - GPS Interface
```

10.4 出现端口类似不是从 ttyUSB0 开始的情况如何处理

需确定ttyUSB口的使用情况。在断开模块时，确认系统是否已释放ttyUSB0端口资源。

10.5 如何手动加载串口驱动

- (1) 确认已配置添加编译驱动serial进入内核（参考章节5 USB串口驱动集成）。
- (2) 查询模块的VID和PID（Linux系统上可通过输入lsusb获取，如果不支持该命令，可以通过windows或相关产品文档获得）。若得到模块的VID=3466，PID=3301，则说明MT571X模块USB串口驱动加载成功。
- (3) 打开Terminal，输入echo "3466 3301" >/sys/bus/usb-serial/drivers/option1/new_id。（这个方法是建立在系统有usbserial模块的基础上，通过下面命令查询“lsmod | grep usbserial*”，如果有，直接使用命令；如果没有，输入命令“modprobe usbserial”进行模块加载）
- (4) 输入dmesg或者ls -all /dev/ttyUSB*，查看加载是否成功。

10.6 若无法映射端口或无法查找对应端口形态需要提供哪些log

- (1) 打开Terminal，执行命令dmesg并保存输出结果于dmesg.txt文档。
- (2) 执行命令ls -l /sys/bus/usb/drivers/，保存输出结果于logcat.txt文档，确保此目录下有option选项。
- (3) 执行命令ls -l /sys/bus/usb/ drivers/option /***/，保存输出结果于logcat.txt文档。
- (4) 执行命令cat /sys/bus/usb/ drivers/option /***/blInterface*，保存输出结果于logcat.txt文档。
- (5) 如果能执行命令cat /proc/bus/usb/devices，将输出结果保存于logcat.txt文档（如果不能请忽略此命令）。

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缩略语

缩略语	英文全名	中文解释
CDC	Communications Device Class	连接设备配置
ECM	Ethernet Networking Control Model	以太网控制模型
LTE	Long Term Evolution	长期演进
OS	Operating system	操作系统
USB	Universal Serial Bus	通用串行总线
NCM	Network Control Model	网络控制模型